

MATH 2101 LINEAR ALGEBRA I, FALL 2025 – ASSIGNMENT 5
OUTLINE OF SOLUTION

Due date: 3rd November (Monday), 2025 10:00pm.

- To receive *full credits*, the solution has to be *clear* and provide sufficient explanations.
- All solutions have to be turned in HKU moodle in the format of *PDF file*.
- Please include your *Name, UID, Faculty, Major (if declared)* in your solution.

1. Let $\mathcal{P}_5$ be the set of polynomials with degree less than or equal to 5. It is known that the set

$$S = \{1 + x - x^2, x + x^2, 3x + 2x^4\}.$$

is linearly independent. What is the dimension of $\mathcal{P}_5$? Find a basis for $\mathcal{P}_5$ which contains S and *other monomials*.

Sol: Note that $\{1, x, x^2, x^3, x^4, x^5\}$ forms a basis for $\mathcal{P}_5$ since any polynomial with degree less than or equal to 5 has the form

$$a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5$$

and the monomials $1, x, x^2, x^3, x^4, x^5$ are linearly independent. (Here 1 means the constant function, x^i means the function $f(x) = x^i$.) Thus, the dimension of $\mathcal{P}_5$ is 6.

Note that $x^4 \notin S$ and so

$$S' = \{1 + x - x^2, x + x^2, 3x + 2x^4, x^4\}$$

is still linearly independent. Similarly, $x^5 \notin S'$ and so

$$S'' = \{1 + x - x^2, x + x^2, 3x + 2x^4, x^4, x^5\}$$

is again linearly independent. Again, $x^3 \notin S''$ and so

$$S''' := \{1 + x - x^2, 3x + 2x^4, x^4, x^5, x^3\}$$

is also linearly independent. Since S''' also has 6 vectors, S''' forms a basis for $\mathcal{P}_5$.

2. Let $V = \{(a_1, a_2) : a_1, a_2 \in \mathbb{R}\}$. Define addition on V as usual: for $(a_1, a_2) \in V$ and $(b_1, b_2) \in V$,

$$(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2 + b_2).$$

Define scalar multiplication on V by: for $c \in \mathbb{R}$ and $(a_1, a_2) \in V$, define

$$c(a_1, a_2) = \begin{cases} (0, 0) & \text{if } c = 0 \\ (\frac{a_1}{c}, ca_2) & \text{if } c \neq 0 \end{cases}$$

Determine whether V is a vector space.

Sol: No, it is not a vector space. Note that

$$(2+2)(a_1, a_2) = 4(a_1, a_2) = \left(\frac{a_1}{4}, 4a_2\right)$$

and

$$2(a_1, a_2) + 2(a_1, a_2) = \left(\frac{a_1}{2}, 2a_2\right) + \left(\frac{a_1}{2}, 2a_2\right) = (a_1, 4a_2).$$

Therefore,

$$(2+2)(a_1, a_2) \neq 2(a_1, a_2) + 2(a_1, a_2)$$

for some non-zero a_1 (and any a_2). This violates the distributive property for the scalar multiplication of a vector space.

3. Let u, v be vectors in a vector space V . Show that if $u + v, u - v$ are linearly independent, then $u + 2v, u - 2v$ are linearly independent.

Sol: Note that

$$u + 2v = \frac{3}{2}(u + v) - \frac{1}{2}(u - v)$$

$$u - 2v = -\frac{1}{2}(u + v) + \frac{3}{2}(u - v).$$

Then, consider the equation:

$$a(u + 2v) + b(u - 2v) = 0.$$

We then have:

$$\left(\frac{3}{2}a - \frac{1}{2}b\right)(u + v) + \left(-\frac{1}{2}a + \frac{3}{2}b\right)(u - v) = 0$$

Since $u + v$ and $u - v$ are linearly independent, we have

$$\begin{aligned} \frac{3}{2}a - \frac{1}{2}b &= 0 \\ -\frac{1}{2}a + \frac{3}{2}b &= 0 \end{aligned}$$

Hence,

$$\begin{pmatrix} \frac{3}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Since the matrix $\begin{pmatrix} \frac{3}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} \end{pmatrix}$ is invertible, we must have

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

to be the only solution. Thus, $u + 2v$ and $u - 2v$ are linearly independent.

4. (a) Let S_1 and S_2 be two sets of linearly independent vectors in a vector space V . Suppose $S_1 \cap S_2 = \emptyset$. Show that $S_1 \cup S_2$ is still linearly independent in V if and only if $\text{span}(S_1) \cap \text{span}(S_2) = \{0\}$.
- (b) Let A_1 and A_2 be two $n \times m$ matrices. Let $d_1 = \dim N(A_1)$ and let $d_2 = \dim N(A_2)$. Show that if $d_1 + d_2 > m$, then there is a non-zero *common* solution in $\mathbb{R}^m$ for the two systems $A_1x = 0$ and $A_2x = 0$.
- (c) What can one say about the previous question if $d_1 + d_2 \leq m$?

Proof. (a) If $S_1 \cup S_2$ is linearly independent in V , then for any $v \in \text{span}(S_1) \cap \text{span}(S_2)$, one can write

$$v = \sum_{i=1}^m a_i v_i = \sum_{j=1}^n b_j w_j, \quad a_i, b_j \in \mathbb{R}, v_i \in S_1, w_j \in S_2.$$

In other words,

$$\sum_{i=1}^m a_i v_i - \sum_{j=1}^n b_j w_j = 0.$$

It follows from the linear independence of $S_1 \cup S_2$ that $b_j = a_i = 0$ for all i, j , i.e., $v = 0$.

Conversely, we assume that $\text{span}(S_1) \cap \text{span}(S_2) = 0$. Then we pick a finite subset $\{v_i\}_{i=1}^m \cup \{w_j\}_{j=1}^n$ of $S_1 \cup S_2$ for some nonnegative integer m, n , where WLOG, $\{v_i\} \subset S_1$ and $\{w_j\} \subset S_2$. Suppose

$$\sum_{i=1}^m a_i v_i - \sum_{j=1}^n b_j w_j = 0$$

for some real numbers a_i and b_j . Then

$$\sum_{i=1}^m a_i v_i = \sum_{j=1}^n b_j w_j \in \text{span}(S_1) \cap \text{span}(S_2) = 0.$$

Linear independence of S_1 and S_2 implies that $a_i = b_j = 0$ for all i, j . It follows that $S_1 \cup S_2$ is a linearly independent subset of V .

(b) By the dimension formula, we know that $\dim(\text{Col}(A_i)) + \dim(N(A_i)) = m$ for $i \in \{1, 2\}$. Then one has

$$\dim(\text{Col}(B)) \leq \dim(\text{Col}(A_1)) + \dim(\text{Col}(A_2)) = 2m - (d_1 + d_2) < m,$$

where $B = \begin{pmatrix} A_1 \\ A_2 \end{pmatrix}$ is a $2n \times m$ matrix. It follows again from the dimension formula that

$$\dim(N(B)) = m - \dim(\text{Col}(B)) > m - m = 0.$$

So there is a nonzero $x \in N(B)$ such that $Bx = 0$, i.e., $A_1 x = A_2 x = 0$.

(c) Nonzero solution in $\mathbb{R}^m$ may or may not exist. Consider $A_1 = (1, 0)^T$ and $A_2 = (0, 1)^T$, we have $d_1 + d_2 = 0 < 1$. Clearly $\dim(N(A_1) \cap N(A_2)) = 0$, i.e., there is no common nonzero solution.

On the other hand, consider $A_1 = A_2 = (1, 0)$, we have $d_1 + d_2 = 1 + 1 = 2$. Note that $x = (0, 1)$ is a nonzero solution for $A_1 x = A_2 x = 0$.

□